

Mitigating Bias in Hiring Decisions through Fair Latent Representation Learning

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Abstract—Machine learning systems used in hiring may inherit historical bias and produce unfair outcomes across demographic groups. This project investigates whether fairness-aware latent representation learning can reduce such bias while preserving predictive utility. A synthetic hiring dataset with controlled bias mechanisms was constructed to model disparities across gender and race. A baseline multilayer perceptron, standalone adversarial debiasing, and ablation variants of a proposed Fair Conditional Variational Autoencoder (Fair CVAE) were evaluated. The proposed model combines adversarial learning, HSIC regularisation, and latent-space projection to reduce sensitive information leakage from the learned representation. Results show that the full Fair CVAE achieves the strongest fairness improvement, increasing minimum intersectional Disparate Impact from 0.34 to 0.70 while maintaining competitive predictive performance. Latent-space analysis further shows large reductions in gender centroid distance and MMD, suggesting that the fairness gains reflect representational change rather than threshold effects alone. These findings indicate that structured latent-space debiasing can reduce dominant sensitive group structure more effectively than single-mechanism adversarial debiasing on this controlled benchmark.

Index Terms—bias mitigation, machine learning, neural networks, representation learning, variational autoencoder

1 INTRODUCTION

1.1 Motivation

MACHINE learning systems are increasingly used in high-stakes domains such as hiring, lending, and criminal justice. In hiring, automated screening tools determine which candidates are reviewed by humans, meaning that any bias in these systems directly affects who progresses in the recruitment process. The technology sector has been repeatedly criticised for uneven demographic representation, and biased screening models risk reinforcing this pattern at scale. This makes automated hiring a concrete technical setting in which diversity in technology can be studied not only as a representation problem, but also as a machine learning fairness problem.

A common first response is to remove the sensitive attribute from the input — drop the *gender* column, drop the *race* column, and retrain. This is usually insufficient. Other features such as referral networks, interview scores, and cultural-fit ratings correlate strongly with demographic group, and a model can recover group membership from them even when the attribute itself is hidden. Removing a feature hides the information; it does not remove it.

A more serious approach is adversarial debiasing, where an auxiliary classifier tries to predict the sensitive attribute from an intermediate representation, and the main model is penalised when it succeeds. Although this can reduce direct predictability, prior work suggests that adversarial training alone may not be sufficient to remove all sensitive information from the learned representation [1], [2]. Residual information may remain in non-linear, distributional, or group-level structure even when a simple adversary becomes less effective.

The risks here are not hypothetical. In 2018, Reuters reported that Amazon had scrapped an internal AI recruiting tool after discovering it systematically downgraded female applicants; the system had been trained on a decade of mostly male resumes and effectively taught itself that male candidates were preferable. It penalised resumes containing the word “women’s” — as in “women’s chess club captain” — and lowered the scores of graduates from two all-women’s colleges. Notably, even after Amazon edited the programs to neutralise these specific terms, engineers could not guarantee that the model would not find other ways of sorting candidates that were equally discriminatory, and the project was eventually abandoned [3]. This case illustrates the central technical problem of this project: once bias is absorbed into the learned representation, removing a few obvious surface markers does not remove the bias itself.

Regulators have begun to respond to the risks of automated employment decision tools. For example, recent legislation in some U.S. jurisdictions now requires employers using such tools to commission independent bias audits, publish the results, and notify candidates that AI-based screening is being used. Against this backdrop, the technical question of how to reduce bias in a hiring model, rather than merely how to measure it, has become directly relevant to both practitioners and regulators.

1.2 Research Question

This project investigates whether fairness can be improved more robustly by learning a more structured and tightly regularised latent representation, rather than relying on standalone adversarial debiasing in a standard discriminative model. The research question is:

Can a structured latent representation, jointly regularised by adversarial training, statistical dependence minimisation, and geometric projection, reduce intersectional bias in hiring decisions more effectively than standalone adversarial debiasing on hidden features, while still maintaining predictive performance?

To answer this, a Fair Conditional Variational Autoencoder (Fair CVAE) is proposed that combines three complementary debiasing mechanisms in a shared latent space. The model is evaluated on a controlled synthetic hiring dataset against a baseline multilayer perceptron, a standalone adversarial classifier, and two ablation variants of the proposed model, together with the complete model.

1.3 Contributions

This paper makes the following contributions:

- A Fair CVAE architecture that combines adversarial learning, HSIC-based dependence minimisation, and orthogonal projection within a shared latent space, together with a gradient-projection step that prevents suppressed sensitive directions from re-emerging during training.
- A four-phase curriculum training schedule that stabilises the joint optimisation of reconstruction, prediction, and three fairness objectives over 350 epochs.
- A controlled synthetic hiring dataset with bias injected across resume, interview, and culture-fit stages, supporting per-group and intersectional evaluation over eight subgroups.
- Experimental results showing that the full model improves minimum intersectional Disparate Impact from 0.34 to 0.70 and reduces the latent centroid distance between gender groups by over 92%, outperforming standalone adversarial debiasing and two ablation variants.

1.4 Paper Structure

The remainder of this paper is organised as follows. Section 2 reviews prior work on algorithmic fairness, adversarial debiasing, and fair representation learning. Section 3 describes the dataset, the baseline models, and the proposed Fair CVAE. Section 4 reports predictive and fairness results, including per-group and latent-space analyses. Section 5 discusses the findings, limitations, and threats to validity. Section 6 concludes and outlines future work.

2 RELATED WORK

This section reviews prior work in four areas relevant to this project. The discussion first considers fairness metrics used to quantify disparity in algorithmic decision-making (Section 2.1), then examines diagnostic tools for identifying the sources of bias (Section 2.2). The remaining two subsections review the two main families of mitigation methods most relevant to this project: adversarial debiasing (Section 2.3) and fair representation learning (Section 2.4).

2.1 Fairness Metrics and Their Trade-offs

A range of fairness criteria have been proposed to quantify disparity in algorithmic decision-making, particularly in high-stakes classification tasks such as hiring. Among the most widely used group-level metrics are Disparate Impact and Equal Opportunity. Disparate Impact [4] compares positive prediction rates across demographic groups and is often evaluated against the “four-fifths rule” used in U.S. employment law, where a ratio below 0.8 is treated as evidence of adverse impact. Equal Opportunity [5] instead compares true positive rates across groups, aiming to ensure that comparably qualified candidates have equal chances of a favourable decision regardless of group membership. These two metrics capture complementary notions of fairness: Disparate Impact focuses on overall outcome parity, while Equal Opportunity conditions on qualification.

These criteria, however, cannot in general be satisfied simultaneously. Chouldechova [6] and Kleinberg et al. [7] show that common fairness definitions are mutually incompatible whenever base rates differ across groups. As a result, fairness evaluation cannot usually rely on a single criterion, and metric choice depends on which aspect of disparity a study aims to prioritise. In this project, Disparate Impact is treated as the primary fairness indicator because of its close alignment with the four-fifths rule and employment-related fairness auditing, while Equal Opportunity is used as a secondary diagnostic. Because qualification is itself difficult to define cleanly in hiring, Equal Opportunity is interpreted more cautiously than Disparate Impact in the later evaluation.

2.2 Bias Analysis and Explainability

This subsection discusses prior work on understanding the sources of bias in machine learning models. The discussion first considers post-hoc explanation methods for attributing model predictions to individual features, then considers controlled statistical analysis used to isolate residual group-level effects after adjusting for observable qualifications.

Ribeiro et al. [8] proposed LIME, a post-hoc explanation method that fits an interpretable surrogate model, typically a linear model, around an individual prediction by perturbing the input locally. Although LIME is intuitive and computationally practical, its reliance on local random perturbations can lead to variation in feature attributions across runs. This is a limitation for this project, because the bias diagnosis stage requires feature attributions to be aggregated across many candidates in order to identify group-level patterns. Unstable local explanations would therefore weaken the reliability of the resulting group-level analysis.

Lundberg and Lee [9] proposed SHAP, which grounds feature attribution in cooperative game theory. SHAP values satisfy properties such as additivity and consistency, making them more suitable for comparing feature contributions across many predictions. While both LIME and SHAP are fundamentally local explanation methods, SHAP’s additive structure allows per-candidate attributions to be aggregated more systematically across demographic groups. For this reason, SHAP is used in Section 3 as the primary feature-attribution tool. In this project, TreeExplainer is applied to

a Gradient Boosting classifier to estimate feature contributions and identify which variables most strongly contribute to group-level prediction gaps.

In addition to feature attribution, controlled logistic regression is used to estimate residual group-level effects after adjusting for observable qualifications. This helps distinguish disparities associated with qualification-related features from disparities that persist after such controls are included. Together, SHAP and controlled logistic regression provide complementary diagnoses: SHAP identifies which features drive group-level prediction gaps, while logistic regression quantifies how much disadvantage remains after observable qualifications are accounted for. These analyses inform the design of the representation-level intervention proposed in this project.

2.3 Adversarial Debiasing

This subsection reviews adversarial approaches to debiasing, in which an auxiliary classifier is trained to recover the sensitive attribute from a model’s representation, while the main model is trained to make this recovery difficult. The discussion first considers the underlying training mechanism, then its application to fairness, and finally its limitations relevant to this investigation.

The technical mechanism most commonly used in modern adversarial debiasing is the Gradient Reversal Layer (GRL), introduced by Ganin and Lempitsky [10]. The GRL acts as the identity function during forward propagation but reverses the sign of the gradient during backpropagation, allowing a feature extractor and an adversarial classifier to be trained jointly. Although originally proposed for domain adaptation, the same mechanism can be repurposed for fairness by treating the sensitive attribute as the adversarial target. This is the mechanism used by both adversarial models in this project.

Zhang et al. [11] provided an influential early application of adversarial learning to fairness, training a predictor together with an adversary that attempts to recover protected information. The predictor is optimised to perform the main task while reducing the adversary’s ability to infer the sensitive attribute. However, adversarial min-max training can be sensitive to imbalance between the predictor and adversary and can be difficult to stabilise in practice. Adel et al. [1] developed this direction further by proposing a one-network adversarial fairness framework, in which the adversarial objective is integrated into a single differentiable classifier. The adversarial baseline implemented in this project follows this integrated GRL-based style, although the full Fair CVAE additionally updates the adversary for a small number of inner steps to improve training stability.

While adversarial methods can reduce direct predictability of the sensitive attribute, they are not always sufficient on their own. Song et al. [2] further motivate representation-level fairness by showing that controlling sensitive information in learned representations is central to fair prediction. This motivates the design of the proposed Fair CVAE in Section 3, which combines adversarial training with complementary representation-level regularisers. The next subsection therefore reviews fair representation learning, which is the broader setting into which the proposed Fair CVAE fits.

2.4 Fair Representation Learning

This subsection reviews fair representation learning, in which the goal is to learn a representation of the input from which downstream predictions are less influenced by the sensitive attribute. The discussion first considers the variational autoencoder framework that underpins many recent methods, then examines two main families of fair representation methods, and finally introduces the statistical-independence regulariser used in the proposed Fair CVAE.

The variational autoencoder (VAE), proposed by Kingma and Welling [12], learns a probabilistic latent representation by jointly optimising a reconstruction objective and a Kullback–Leibler regulariser that encourages the encoder distribution to remain close to a unit Gaussian prior. This variational bottleneck provides a more controllable latent space than standard autoencoders and forms the basis of several fair representation methods. Sohn et al. [13] extended this framework to the conditional setting, in which additional information can be passed to the encoder, decoder, or both. The proposed Fair CVAE follows this conditional design, but restricts the conditioning to the decoder, so that the sensitive attribute does not enter the encoder and is not required by the latent representation for reconstruction.

Within this representation-learning paradigm, two main families of fairness-aware methods are particularly relevant. Louizos et al. [14] proposed the Variational Fair Autoencoder, which augments a variational autoencoder with a Maximum Mean Discrepancy penalty to encourage similarity of latent distributions across sensitive groups. Madras et al. [15] proposed LAFTR, which combines an autoencoder with an adversarial penalty to learn fair and transferable representations. Both works demonstrate that representation-level interventions can reduce group-level disparities, but they apply different fairness pressures: VFAE relies primarily on distribution matching, while LAFTR relies primarily on adversarial learning. The proposed Fair CVAE differs by combining a conditional variational latent model with adversarial training, HSIC-based dependence minimisation, and latent-space projection within a single framework.

A complementary regulariser used in this project is the Hilbert–Schmidt Independence Criterion (HSIC) [16], a kernel-based statistical measure of dependence between two variables. Minimising HSIC between the latent representation and the sensitive attribute encourages statistical independence beyond what an adversary can capture, since HSIC is sensitive to non-linear and distributional structure rather than only direct predictability. The proposed Fair CVAE incorporates HSIC alongside adversarial training and an additional latent-space projection step, with the aim of addressing residual sensitive structure that single-mechanism methods such as VFAE or LAFTR may leave in the representation.

3 METHODOLOGY

3.1 System Overview

The proposed system follows a four-stage pipeline for analysing and mitigating bias in automated hiring decisions.

In the first stage, a synthetic hiring dataset is generated with controlled bias mechanisms embedded across multiple evaluation stages including resume screening, technical interviews, and culture fit assessment. The dataset comprises 15,000 candidates across 8 intersectional groups (2 genders $\times$ 4 races), with 11 candidate features used as input.

In the second stage, systematic bias diagnosis is conducted using fairness metrics, SHAP analysis, and controlled regression to identify the sources and severity of bias in the hiring pipeline.

In the third stage, a baseline classifier is trained to establish reference performance. This is a two-layer MLP (128 $\rightarrow$ 64 $\rightarrow$ 1) with LeakyReLU, BatchNorm, and Dropout, trained with binary cross-entropy loss and no fairness constraints.

In the fourth stage, the Fair Conditional VAE is developed and evaluated. Two ablation variants of the Fair CVAE (*adv_only* and *no_adv*) are compared together with the complete *full* model. Feature removal was considered as a simple conceptual alternative, but was not adopted as a primary experimental baseline because removing bias-associated variables may also discard task-relevant information while leaving proxy information in correlated features.

The dataset is split into training (68%), validation (12%), and test (20%) sets using stratified sampling to preserve group proportions. All experiments use fixed random seeds for reproducibility.

At the software level, the implementation is organised into four modules. The data module generates or loads the hiring dataset and applies preprocessing. The analysis module computes fairness metrics, SHAP explanations, and controlled regression diagnostics. The modelling module trains the baseline MLP, standalone adversarial classifier, and Fair CVAE variants. The web module wraps the same analysis and training pipeline in a FastAPI interface, allowing CSV upload, model selection, live job monitoring, and result visualisation.

3.2 Bias Diagnosis

Prior to applying debiasing methods, a systematic bias diagnosis is conducted to identify the sources and extent of bias in the hiring pipeline. This diagnostic stage informs the design of the subsequent debiasing model by identifying which types of bias, namely group-level disparity, feature-level contribution, and residual effects after controlling for qualifications, must be addressed.

3.2.1 Fairness Metrics

Two group fairness metrics are used to quantify bias across intersectional groups (gender $\times$ race, yielding 8 groups).

Disparate Impact (DI) is measured as the ratio of each group's hiring rate relative to the reference group, Male_White:

$$DI_g = \frac{P(\hat{Y} = 1 \mid G = g)}{P(\hat{Y} = 1 \mid G = \text{Male_White})}$$

A DI value below 0.8 is considered evidence of adverse impact under the four-fifths rule. However, DI alone does not account for qualification differences, motivating the use of Equal Opportunity (EO) to ensure fairness among qualified candidates.

Because the dataset does not contain an externally validated ground-truth qualification label, Equal Opportunity is treated as a proxy metric in this project. A proxy qualified subset is constructed from observed qualification-related features, and the proxy EO gap measures differences in positive prediction rates within this subset. This provides an approximate diagnostic of whether candidates with comparable observed qualifications receive similar hiring opportunities across demographic groups, but it should not be interpreted as a definitive measure of true Equal Opportunity.

3.2.2 SHAP Analysis

The SHAP analysis is used as an attribution-based diagnostic rather than as causal evidence: it indicates which features are associated with group-level prediction gaps in the trained model, but does not by itself prove that these features causally produce the disparities. A Gradient Boosting classifier is trained on the full feature set, and Shapley values are computed for each prediction.

For each feature, the average SHAP value difference between demographic groups is computed. Features with large group-level SHAP differences indicate potential sources of bias. This analysis helps distinguish between proxy variables that encode sensitive information and genuinely predictive features, which is critical for designing effective debiasing strategies.

3.2.3 Controlled Analysis

Logistic regression with controlled variables is used to estimate odds ratios for demographic group indicators while holding observable qualification-related features constant. This allows the project to estimate residual group-level associations after observable qualifications are controlled for, distinguishing disparities associated with qualification-related features from disparities that persist after those controls are included.

Together, these analyses provide a multi-level diagnosis of bias, covering outcome disparities, feature-level contributions, and residual effects beyond observable qualifications.

3.3 Baseline Models

To provide meaningful points of comparison for the proposed Fair Conditional Variational Autoencoder (Fair CVAE), two baseline models are used. The first is a standard multilayer perceptron (MLP) trained only for predictive performance, without any explicit fairness intervention. The second is a simpler adversarial debiasing baseline that applies fairness pressure through a Gradient Reversal Layer (GRL), but does not use generative modelling or latent-space regularisation.

3.3.1 Baseline MLP

A baseline classifier is first trained in order to establish reference predictive performance and fairness behaviour in the absence of any debiasing mechanism. This model is implemented as a two-hidden-layer multilayer perceptron with architecture

$$11 \rightarrow 128 \rightarrow 64 \rightarrow 1,$$

where the input consists of the 11 candidate features used throughout the study. Each hidden layer uses LeakyReLU activation, Batch Normalisation, and Dropout, followed by a sigmoid output unit for binary hiring prediction. The model is trained using binary cross-entropy loss with the AdamW optimiser.

The purpose of this baseline is to represent a standard predictive model that prioritises classification performance alone. It provides a reference point against which subsequent fairness-aware methods can be compared in terms of both predictive utility and group disparity. In particular, this baseline is used to quantify the extent to which the original hiring pipeline reproduces adverse impact across demographic groups before any mitigation is applied. The implementation corresponds to the simple classifier used in the experimental codebase.

3.3.2 Standalone Adversarial Baseline

A standalone adversarial debiasing classifier is implemented as a primary fairness-aware baseline. This model provides the main comparison against which the proposed Fair CVAE is evaluated, because adversarial debiasing is one of the most common representation-level fairness interventions. For brevity, this baseline is referred to as Standalone GRL in the results.

The model uses a shared feature extractor, a prediction head for hiring classification, and a GRL-based adversarial head for sensitive-attribute prediction. During training, the Gradient Reversal Layer encourages the feature extractor to produce hidden representations that are predictive of hiring outcomes but less informative about the sensitive attribute. Unlike the Fair CVAE adversary, this baseline uses a plain MLP adversary without spectral normalisation.

Unlike the proposed Fair CVAE, this baseline does not include reconstruction, KL regularisation, HSIC dependence minimisation, orthogonal projection, or gradient projection. It therefore isolates the effect of adversarial pressure in a standard discriminative model. Two versions are evaluated: one using gender as the adversarial target and one using intersectional group membership as the adversarial target. The GRL strength α is held at 0 for the first 80 of 200 training epochs to allow the predictor to stabilise, then increased smoothly to 1.0 using a sigmoid schedule. The total training loss is $\mathcal{L}_{\text{pred}} + 1.5 \mathcal{L}_{\text{adv}}$, optimised using Adam.

3.4 Fair Conditional Variational Autoencoder

3.4.1 Model Overview

To mitigate bias while preserving predictive utility, this project proposes a Fair Conditional Variational Autoencoder (Fair CVAE), based on variational and conditional autoencoder principles [12], [13], which combines probabilistic representation learning with multiple fairness constraints. The aim is not to completely eliminate all sensitive information from the latent space, which is often unrealistic in practice, but rather to reduce dominant group-level structure and limit the extent to which sensitive attributes remain actionable for downstream hiring prediction.

The model consists of four components: an encoder, a decoder, a predictor, and an adversary. The encoder maps the input feature vector x to a latent representation z ,

the decoder reconstructs the input conditioned on both z and the sensitive attribute a , the predictor estimates the hiring decision from the latent representation alone, and the adversary attempts to recover the sensitive attribute from the latent space.

This structure allows the model to explicitly account for sensitive information during reconstruction, while discouraging its use in the final prediction. The overall architecture is illustrated in Figure 1.

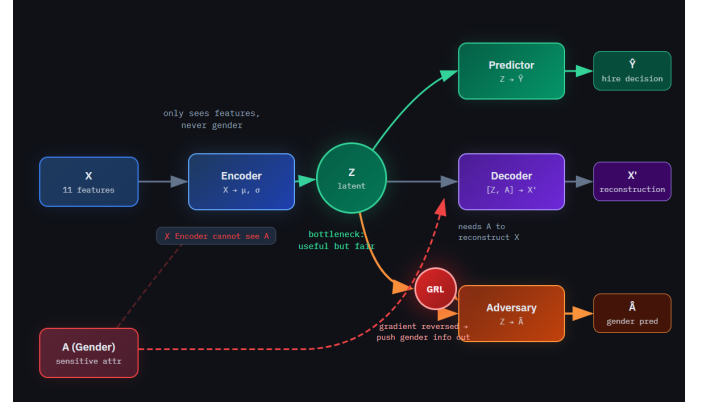


Fig. 1. Architecture of the Fair CVAE model. The encoder maps input features X to latent representation Z without observing the sensitive attribute A . The predictor outputs hiring decisions from Z alone. The decoder reconstructs X conditioned on A . The adversary, connected through a Gradient Reversal Layer (GRL), encourages Z to contain less recoverable gender information.

Formally, let $x \in \mathbb{R}^d$ denote the input feature vector, $y \in \{0, 1\}$ the hiring label, and a the sensitive attribute (gender in this project). The model learns a latent representation $z \in \mathbb{R}^k$ such that:

$$q_\phi(z | x) = \mathcal{N}(z; \mu_\phi(x), \text{diag}(\sigma_\phi^2(x))), \quad (1)$$

where $\mu_\phi(x)$ and $\sigma_\phi^2(x)$ are produced by the encoder.

This design is motivated by the limitation of standard adversarial debiasing methods, which often reduce direct leakage of sensitive information but may still leave residual group structure in the latent space. By combining adversarial training, dependence minimisation, and geometric projection, the proposed model aims to produce a more robust fair representation.

3.4.2 Model Architecture

The encoder is implemented as a fully connected neural network with dimensions

$$11 \rightarrow 256 \rightarrow 256 \rightarrow 128 \rightarrow (\mu, \log \sigma^2),$$

where the latent dimensionality is set to 64. Each hidden layer uses LeakyReLU activation, Layer Normalisation, and Dropout. These choices improve optimisation stability and help prevent overfitting.

The encoder outputs the latent mean $\mu \in \mathbb{R}^{64}$ and log-variance $\log \sigma^2 \in \mathbb{R}^{64}$. To ensure numerical stability, the log-variance is clamped to the range $[-10, 2]$. Sampling is performed using the reparameterisation trick:

$$z = \mu + \epsilon \odot \sigma, \quad \epsilon \sim \mathcal{N}(0, I), \quad (2)$$

where $\sigma = \exp(0.5 \log \sigma^2)$.

The decoder reconstructs the original feature vector from the latent representation and the sensitive attribute:

$$\hat{x} = f_{\text{dec}}(z, a). \quad (3)$$

Conditioning the decoder on a is important because it reduces the pressure on the latent representation to encode sensitive information purely for reconstruction purposes. This helps separate the goals of faithful reconstruction and fair prediction.

The predictor receives only the latent representation and outputs a hiring logit:

$$\hat{y} = f_{\text{pred}}(z). \quad (4)$$

This means that the final decision is made from the learned representation rather than directly from the sensitive attribute.

An adversary is attached to the latent representation in order to predict the sensitive attribute:

$$\hat{a} = f_{\text{adv}}(z). \quad (5)$$

To make the adversarial signal more stable, the adversary uses spectral normalisation and Dropout. A class-weighted cross-entropy loss is used to handle class imbalance in the sensitive attribute, and the adversary is periodically reset during training to avoid collapse or overfitting.

All linear layers are initialised using Xavier initialisation. Separate AdamW optimisers are used for the main model and the adversary, with cosine annealing learning-rate schedules applied throughout training.

3.4.3 Fairness Mechanisms

Although adversarial training can reduce direct sensitive leakage, it is often insufficient on its own. Therefore, three complementary fairness mechanisms are incorporated.

First, adversarial debiasing is implemented using a Gradient Reversal Layer (GRL) [10]. During forward propagation the GRL behaves as the identity function, but during backpropagation it multiplies the incoming gradient by $-\alpha$, thereby encouraging the encoder to learn features that are uninformative for predicting the sensitive attribute. This can be written as:

$$\text{GRL}(z) = z, \quad (6)$$

with backward gradient

$$\frac{\partial \text{GRL}(z)}{\partial z} = -\alpha I. \quad (7)$$

Second, Hilbert-Schmidt Independence Criterion (HSIC) [16] regularisation is used to minimise statistical dependence between the latent representation z and the sensitive attribute a . Given kernel matrices K and L for z and a , respectively, empirical HSIC is defined as:

$$\text{HSIC}(Z, A) = \frac{1}{(n-1)^2} \text{tr}(KHLH), \quad (8)$$

where $H = I - \frac{1}{n} \mathbf{1}\mathbf{1}^T$ is the centering matrix. In this project, an RBF kernel is used for the latent representation, with bandwidth selected using the median heuristic, while the sensitive attribute is represented through one-hot encoding. To reduce computational cost, HSIC is estimated on at most 512 samples per batch.

Third, orthogonal latent projection is used to remove directions in latent space that are most predictive of the sensitive attribute. These directions are identified by fitting logistic regression probes with multiple regularisation strengths and stacking their weight vectors. Singular Value Decomposition (SVD) is then applied to extract the dominant sensitive directions. If d_i denotes such a direction, the projected latent representation is:

$$z_{\text{clean}} = z - \sum_{i=1}^m \langle z, d_i \rangle d_i. \quad (9)$$

This projection removes the most salient linear components associated with the sensitive attribute.

In addition, gradient projection is applied during training to prevent the model from reintroducing sensitive information into the latent mean layer. Specifically, gradients at the latent mean output are projected to remove components aligned with the estimated sensitive directions. This acts as an optimisation-level fairness control that complements both HSIC regularisation and latent projection.

Together, these three main mechanisms operate at different levels: adversarial learning reduces direct predictability, HSIC reduces statistical dependence, and projection removes dominant sensitive geometry. Their combination forms the core debiasing strategy of the proposed model.

TABLE 1
Summary of Fairness and Stabilisation Mechanisms

Mechanism	Level	Target
GRL Adversary	Training	Reduce recoverable A info in Z
HSIC	Statistical	$Z \perp A$
Orthogonal Proj.	Geometric	Remove sensitive dirs
Gradient Proj.	Optimisation	Prevent re-encoding
KL Annealing	Regularisation	Prevent collapse

3.4.4 Objective Function

The overall training objective combines reconstruction, variational regularisation, predictive performance, adversarial fairness, and HSIC dependence minimisation:

$$\mathcal{L} = \mathcal{L}_{\text{rec}} + \beta \mathcal{L}_{\text{KL}} + \mathcal{L}_{\text{pred}} + \lambda_{\text{adv}} \mathcal{L}_{\text{adv}} + \lambda_{\text{hsic}} \mathcal{L}_{\text{hsic}}. \quad (10)$$

The reconstruction loss encourages faithful decoding of the original feature vector:

$$\mathcal{L}_{\text{rec}} = \|x - \hat{x}\|_2^2. \quad (11)$$

The KL divergence regularises the approximate posterior towards a unit Gaussian prior:

$$\mathcal{L}_{\text{KL}} = -\frac{1}{2} \sum_{j=1}^k (1 + \log \sigma_j^2 - \mu_j^2 - \sigma_j^2). \quad (12)$$

To avoid posterior collapse, KL annealing is applied, and a free-bits mechanism enforces a minimum KL contribution per latent dimension. This prevents the KL term from forcing the latent variables to become uninformative too early in training.

The prediction loss is binary cross-entropy on the hiring label:

$$\mathcal{L}_{\text{pred}} = -(y \log \hat{y} + (1 - y) \log(1 - \hat{y})). \quad (13)$$

The adversarial loss is a class-weighted cross-entropy objective for predicting the sensitive attribute:

$$\mathcal{L}_{\text{adv}} = - \sum_c w_c a_c \log \hat{a}_c, \quad (14)$$

where w_c denotes the class weight for sensitive class c . The term a_c denotes the one-hot sensitive-class label, and $\hat{a}_c$ denotes the adversary’s predicted probability for class c .

Although the adversarial loss appears with a positive sign in the overall objective, the Gradient Reversal Layer reverses its gradient with respect to the encoder, so that minimising this objective still encourages the latent representation to become less informative about the sensitive attribute.

Balancing these terms is essential. Excessively strong fairness regularisation may reduce reconstruction quality or predictive accuracy, while insufficient fairness pressure may leave the latent representation highly sensitive to gender.

3.4.5 Training Strategy

Training is carried out in four phases over 350 epochs in order to improve optimisation stability. The schedule described below applies to the full Fair CVAE. For ablation variants, the same epoch structure is retained, but disabled components have zero weight according to the ablation setting.

In Phase 1 (epochs 1–50), the model focuses on establishing a useful predictive signal. Fairness constraints are disabled, with $\beta = 0$, adversarial strength $\alpha = 0$, and no projection applied. This avoids destabilising the representation before it has learned meaningful task-related structure.

In Phase 2 (epochs 51–150), reconstruction, KL regularisation, HSIC regularisation, and adversarial training are introduced gradually. The KL weight is annealed from 0 to 0.1, while the GRL strength increases from 0 to 1.0. This smooth transition improves stability and avoids abrupt shifts in the optimisation landscape.

In Phase 3 (epochs 151–250), the full fairness pipeline is activated. Orthogonal projection and gradient projection are enabled, the adversarial signal is strengthened, and HSIC regularisation is increased. The KL weight is further increased from 0.1 to 0.15, and the GRL strength grows towards its maximum value of 8.0.

In Phase 4 (epochs 251–350), the model is fine-tuned with stronger fairness emphasis. The HSIC weight is doubled, while the reconstruction weight is reduced, allowing the model to refine fairness properties without overly prioritising perfect reconstruction.

The adversary is reset every 40 epochs to avoid overfitting to a stale latent representation. The orthogonal projection layer is re-fitted every 20 epochs to ensure that the estimated sensitive directions remain aligned with the current latent space.

The adversary is updated multiple times per iteration to maintain sufficient pressure against sensitive leakage. In the reported runs, key hyperparameters were fixed as follows: latent dimensionality $z = 64$, hidden dimensionality 256, $\lambda_{\text{hsic}} = 50$, $\lambda_{\text{adv}} = 2$, maximum GRL strength $\alpha_{\text{max}} = 8$, free-bits threshold 0.1 per latent dimension, and three sensitive directions removed by orthogonal projection. The adversary is updated for up to five inner steps per

main optimisation iteration, with the number of inner steps introduced gradually during Phase 2 and kept at five during Phases 3 and 4. The training schedule is visualised in Figure 2.

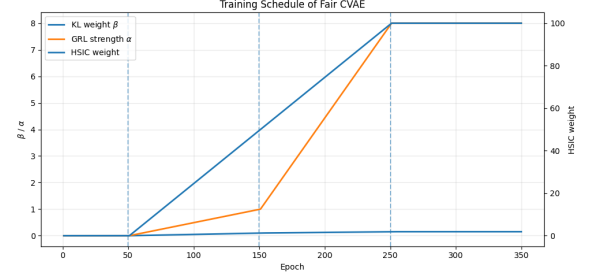


Fig. 2. Training schedule of the Fair CVAE. The KL weight β , GRL strength α , and HSIC weight are gradually increased across four phases (separated by dashed lines). Phase 1 (1–50) performs prediction-only warm-up without fairness or reconstruction losses. Phase 2 (51–150) introduces fairness constraints gradually. Phase 3 (151–250) activates the full pipeline including projection. Phase 4 (251–350) fine-tunes with doubled HSIC weight.

3.4.6 Ablation and Calibration

To assess the contribution of different fairness components, two ablation variants (*adv_only* and *no_adv*) are evaluated alongside the complete *full* model. The *adv_only* setting isolates adversarial debiasing by keeping the adversarial objective while disabling the VAE reconstruction loss, KL regularisation, HSIC regularisation, orthogonal projection, and gradient projection. The *no_adv* setting keeps the VAE reconstruction loss, KL regularisation, and HSIC regularisation, but disables the adversarial objective, adversary updates, orthogonal projection, and gradient projection. The *full* setting includes the complete fairness pipeline, combining the variational bottleneck, HSIC regularisation, adversarial training, orthogonal projection, and gradient projection.

Prediction thresholds are calibrated on the validation set rather than being fixed at 0.5. This avoids introducing arbitrary decision bias through threshold selection and enables fairer comparison across model variants. The test set is reserved for final performance and fairness reporting only.

3.4.7 Representation Analysis

In addition to standard performance and fairness metrics, the learned latent representations are analysed directly in order to assess whether sensitive information remains encoded after debiasing.

For the Fair CVAE, the encoder mean μ was used as the primary latent representation for quantitative analysis. Projection-based visualisations were examined separately for qualitative comparison.

Three quantitative latent-space metrics are used. First, centroid distance measures the Euclidean distance between the mean latent vectors of the two sensitive groups:

$$D_{\text{centroid}} = \|\bar{z}_{\text{male}} - \bar{z}_{\text{female}}\|_2. \quad (15)$$

A smaller value indicates weaker group-level separation.

Second, probe accuracy measures the degree to which a linear classifier can recover the sensitive attribute from the

latent representation. A logistic regression probe is trained on the latent representation, and lower accuracy indicates reduced recoverable sensitive information.

Third, an approximate Maximum Mean Discrepancy (MMD) score is used to estimate the distributional gap between the latent representations of the two sensitive groups. Lower MMD indicates greater overlap between the group distributions.

To complement these quantitative metrics, t-SNE [17] and PCA visualisations are also generated. These plots provide an intuitive qualitative view of whether male and female samples, or intersectional groups, form visibly distinct clusters in latent space. For the baseline model, hidden representations from the final hidden layer are used for comparison, while for the Fair CVAE the encoder mean is used as the latent representation.

3.4.8 Design Rationale

The final design was motivated by the limitations observed in simpler mitigation strategies. A standalone adversarial baseline was included because adversarial debiasing is a widely used fairness intervention and provides a natural point of comparison. However, in preliminary experiments and in the final results, adversarial pressure alone improved fairness only partially. It reduced some direct predictability of the sensitive attribute, but did not sufficiently remove group structure from the learned representation, particularly for the most disadvantaged intersectional subgroups.

A CVAE was chosen instead of a standard GRL-based discriminative model because the project required a more structured representation than an ordinary hidden feature space could provide. In the standalone GRL baseline, adversarial pressure is applied to intermediate hidden features, but those features are not explicitly regularised as a probabilistic latent space. By contrast, the CVAE parameterises the representation through the encoder mean and log-variance, introducing a variational bottleneck and KL regularisation that encourage a more compact and controllable latent structure. This provides a clearer basis for combining adversarial debiasing, HSIC-based dependence minimisation, and projection-based removal of sensitive directions within a shared latent space.

A feature-removal comparison was also considered because removing bias-associated variables is a common intuitive response in hiring-related fairness problems. However, this approach has two important limitations. First, it depends on prior knowledge of which features should be removed. Second, removing selected features does not eliminate proxy information that may still be encoded in correlated variables. Moreover, such removal may also discard information that remains genuinely useful for prediction. For these reasons, the project focused on learning a fairer representation rather than relying on manual feature exclusion as a primary mitigation strategy.

The four-phase curriculum schedule was introduced for optimisation reasons. The full model combines prediction, reconstruction, KL regularisation, adversarial pressure, HSIC minimisation, and projection-based interventions. Enabling all components from the beginning made training less stable and risked weakening the predictive signal before a useful latent structure had formed. The phased schedule

therefore allowed the model to first learn a task-relevant representation and then gradually apply stronger fairness constraints.

Conditioning the decoder on the sensitive attribute was intended to reduce the need for the latent representation itself to retain sensitive information for reconstruction. Without this conditioning, the encoder would be under greater pressure to preserve demographic information in latent space in order to reconstruct the input accurately. By allowing the decoder to access the sensitive attribute directly, reconstruction and fairness objectives could be separated more cleanly.

Overall, the model was designed as a layered mitigation strategy rather than a single-mechanism intervention. The aim was not to claim complete removal of all sensitive information, but to reduce the extent to which sensitive structure remained actionable in the learned representation and downstream hiring predictions.

3.5 Implementation

The full experimental pipeline was implemented in Python. PyTorch was used to develop all neural network models, including the baseline classifier, the alternative adversarial baseline, and the proposed Fair CVAE, because it provides sufficient flexibility for custom training procedures, multi-objective optimisation, and gradient-based fairness interventions. In particular, it enabled direct implementation of the Gradient Reversal Layer, orthogonal projection in latent space, and gradient projection for suppressing sensitive directions during optimisation. NumPy and Pandas were used for data handling and preprocessing, while scikit-learn was used for feature standardisation, logistic regression probes, and Gradient Boosting models for SHAP analysis. SHAP was selected because it provides an interpretable way to identify which input features contribute most to group-level prediction disparities, thereby supporting both the diagnostic stage and the design of comparison baselines.

3.6 Interactive Web Application

To make the proposed pipeline reproducible and to support interactive demonstration, a FastAPI-based web application was developed around the experimental codebase. The application exposes four pages: a home page, an intermediate fairness analysis page, an interactive demo page, and a results page. Through the demo page, a user can upload a CSV dataset, select one of the evaluated models, including the baseline MLP, the gender-based standalone adversarial baseline, the intersectional standalone adversarial baseline, or one of the Fair CVAE modes, and trigger a training run as a background job. A lightweight job manager handles queuing, status polling, and live log streaming, so users can monitor training progress without blocking the web server. Once the job completes, fairness metrics, per-group Disparate Impact, and, for the Fair CVAE modes, latent-space visualisations are rendered directly in the browser. The web application reuses the same underlying data-loading, training, threshold-calibration, evaluation, and visualisation modules as the offline experiments, providing an interactive wrapper around the project pipeline. Jinja2 templates,

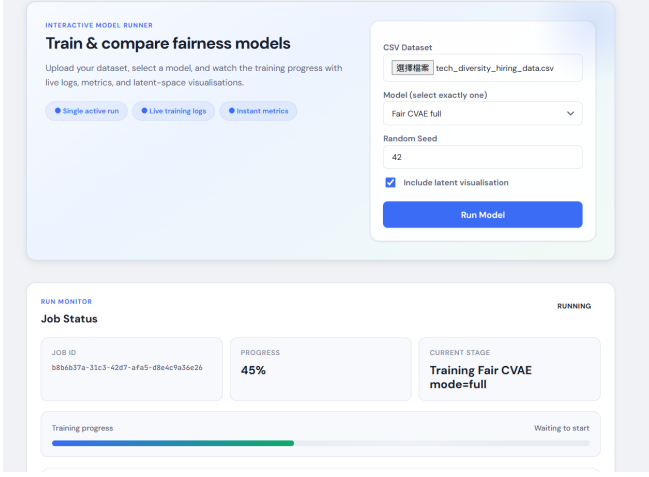


Fig. 3. Interactive web application for running the fairness pipeline. Users can upload a CSV dataset, select a model, monitor training progress, and inspect predictive and fairness metrics after completion.

vanilla JavaScript, and CSS are used on the frontend; no heavyweight JavaScript framework is required.

The web interface was used as a user-facing wrapper around the same offline experimental pipeline, rather than as a separate modelling implementation.

3.7 Experimental Protocol

To ensure fair and reproducible comparison within the main Fair CVAE pipeline, the baseline MLP and Fair CVAE ablation variants were evaluated using the same data split and validation-based threshold calibration procedure. The validation set was used for threshold calibration and for monitoring optimisation behaviour, while the test set was reserved strictly for final reporting. This separation reduces evaluation bias and prevents leakage of information from the test set into model development.

The standalone GRL baselines were trained and evaluated separately using a fixed 0.5 decision threshold. Therefore, comparisons involving the standalone GRL baselines should be interpreted as indicative comparisons under closely related, but not identical, evaluation settings.

Because the dataset does not provide an externally validated qualification label, Equal Opportunity was approximated using a proxy qualified subset constructed from experience, education, algorithmic skill, and interview-related features. This allowed the project to assess not only predictive performance, but also whether fairness interventions improved opportunity for candidates with comparable observable qualifications.

4 RESULTS

4.1 Experimental Setup

All experiments were conducted on the synthetic hiring dataset described in Section 3.1, comprising 15,000 candidates across 8 intersectional groups defined by gender (Male, Female) and race (White, Asian, Hispanic, Black). The dataset was split into training (68%), validation (12%),

and test (20%) sets using stratified sampling over the intersectional group labels. All reported results are on the held-out test set (3,000 candidates). Prediction thresholds for all Fair CVAE variants were calibrated on the validation set to match the baseline positive prediction rate (approximately 30%), ensuring that differences in fairness metrics reflect genuine changes in prediction behaviour rather than shifts in overall prediction volume.

The following methods are compared: (1) the baseline MLP with no fairness intervention, (2) a standalone adversarial debiasing classifier using GRL with gender-based and intersectional adversarial objectives, and (3) two ablation variants of the proposed Fair CVAE (*adv_only* and *no_adv*), together with the complete *full* model. Fairness is assessed using Disparate Impact (DI) across all eight intersectional groups, with the minimum DI used as the primary fairness indicator. Predictive performance is measured using accuracy and F1 score. Latent-space analysis focuses on gender as the sensitive attribute, using centroid distance, probe accuracy, and approximate MMD; intersectional fairness is assessed separately through the group-level DI metrics. Proxy Equal Opportunity results followed the same broad pattern as Disparate Impact, but are not used as the primary ranking criterion because the qualified subset is constructed from observed features rather than from an externally validated qualification label.

4.2 Baseline Bias Diagnosis

These analyses were designed to capture bias at the outcome, feature, and controlled-effect levels respectively.

The baseline MLP achieved an accuracy of 0.6903 and an F1 score of 0.5492 on the test set. Despite reasonable predictive performance, the model exhibited substantial fairness violations. The minimum Disparate Impact was 0.3421 (Female_Black), far below the 0.80 threshold established by the four-fifths rule. Six of the eight groups fell below this threshold, indicating widespread adverse impact across both gender and racial lines.

SHAP analysis suggested that OverallInterviewScore was the largest feature-level contributor to group-level prediction disparity in the diagnostic Gradient Boosting model, with an average SHAP difference of -0.378 across disadvantaged groups. CultureFitScore (-0.052) and HasReferral (-0.045) were the second and third largest contributors. These features appear to act as proxy variables associated with demographic group membership in the simulated hiring pipeline, and they are strongly associated with prediction gaps beyond what is captured by the observed qualification-related controls. Controlled logistic regression confirmed that group membership significantly influenced hiring outcomes even after holding qualifications constant. Female_Black candidates had an odds ratio of 0.60 relative to Male_White, meaning their odds of being hired at equivalent qualification levels were approximately 40% lower. Female_Hispanic (OR = 0.66), Male_Black (OR = 0.69), and Male_Hispanic (OR = 0.71) also exhibited significant residual disadvantage. These results confirm that bias in the baseline model is structural rather than a simple product of qualification differences, motivating the need for multi-constraint fairness intervention on the learned representation.

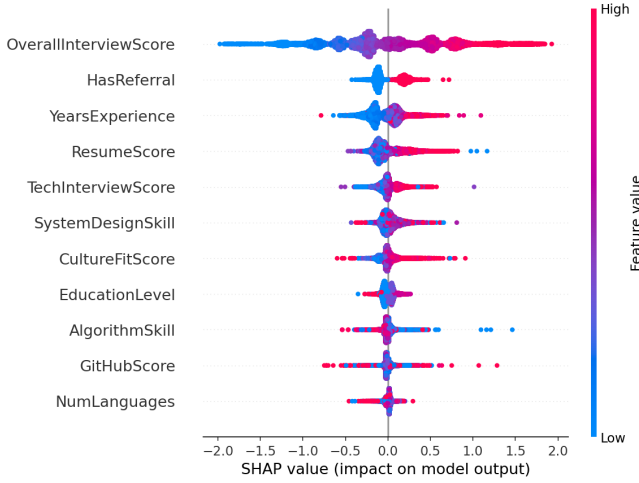


Fig. 4. SHAP summary plot for the baseline model. OverallInterviewScore has the strongest influence on hiring predictions, while HasReferral, ResumeScore, TechInterviewScore, and CultureFitScore also contribute substantially. The distribution of SHAP values also shows how higher or lower feature values shift model output.

4.3 Comparison of Debiasing Methods

Table 2 summarises the predictive performance and fairness outcomes of each method evaluated in this study.

TABLE 2
Performance and Fairness Comparison

Method	Acc	F1	Min DI
Baseline MLP	0.6903	0.5492	0.3421
Standalone GRL (Gender)	0.6807	0.5523	0.4664
Standalone GRL (Intersect.)	0.6763	0.5431	0.4968
CVAE <i>adv_only</i>	0.6513	0.4942	0.6131
CVAE <i>no_adv</i>	0.6617	0.5085	0.4817
CVAE <i>full</i>	0.6597	0.5163	0.7011

The baseline MLP produced the highest accuracy but the worst fairness outcome (Min DI = 0.34). The standalone adversarial baselines using GRL improved fairness moderately: gender-only adversarial training raised Min DI to 0.47, while intersectional adversarial training reached 0.50. However, both remained well below the 0.80 threshold, suggesting that single-mechanism adversarial pressure on hidden representations provides only partial mitigation, even when the adversarial target is extended from gender to intersectional groups.

Among the Fair CVAE variants, the *full* model achieved the strongest fairness outcome (Min DI = 0.70) with an accuracy of 0.6597 and F1 of 0.5163. This more than doubles the minimum DI from 0.3421 to 0.7011, while reducing the gap to parity from 0.6579 to 0.2989. This comes at the cost of a 3.1 percentage point reduction in accuracy and a 3.3 point reduction in F1. The *adv_only* variant achieved Min DI of 0.61 but suffered a larger accuracy and F1 drop, while the *no_adv* variant retained higher accuracy (0.66) but achieved weaker fairness improvement (Min DI = 0.48), comparable to the standalone GRL baselines. This confirms that adversarial training contributes meaningfully to fairness but is insufficient on its own, and that the combination of all mechanisms produces the most effective trade-off between predictive utility and group fairness.

4.4 Per-Group Fairness Analysis

Table 3 presents the per-group Disparate Impact for the baseline and the full Fair CVAE to show how fairness improvements are distributed across intersectional groups.

TABLE 3
Per-Group Disparate Impact: Baseline vs Full CVAE

Group	Baseline	Full CVAE
Male_White	1.0000	1.0000
Male_Asian	0.8923	0.9597
Female_Asian	0.7360	1.0417
Female_White	0.6828	1.0524
Male_Hispanic	0.5847	0.9447
Female_Hispanic	0.5547	0.9492
Male_Black	0.4787	0.8153
Female_Black	0.3421	0.7011

The full Fair CVAE improved DI for all seven disadvantaged groups. The largest gains were observed for Female_Hispanic (0.55 → 0.95), Male_Hispanic (0.58 → 0.94), and Female_Asian (0.74 → 1.04). After debiasing, seven of the eight groups exceeded the 0.80 threshold, compared to only two (Male_White as the reference group, and Male_Asian) in the baseline.

However, Female_Black remained the most disadvantaged group (DI = 0.70), and Male_Black also fell only slightly above the threshold (DI = 0.82). This pattern suggests that intersectional disadvantage involving Black candidates is the most resistant to mitigation under the current approach, possibly because these groups experience the largest cumulative bias across multiple stages of the hiring pipeline simultaneously.

4.5 Latent Space Analysis

To assess whether the observed fairness improvements reflect genuine changes in the learned representation rather than surface-level threshold effects, three quantitative metrics were computed on the latent space. Results across the three CVAE ablation variants are shown in Table 4.

TABLE 4
Latent Space Metrics Across Ablation Variants (Baseline: Centroid Dist. = 1.3713, Probe Acc. = 0.7160, MMD = 0.1416)

Metric	<i>adv_only</i>	<i>no_adv</i>	<i>full</i>
Centroid Dist.	0.8405	0.2860	0.1047
Probe Acc.	0.7000	0.7107	0.7007
MMD	0.0782	0.0258	0.0097

The full model reduced centroid distance by 92.4% and MMD by 93.2% relative to the baseline, indicating that the mean representations and distributional overlap of male and female candidates became substantially more similar in the learned latent space. The *adv_only* variant achieved only moderate reductions (38.7% and 44.8%), while *no_adv* performed better on centroid distance (79.1%) and MMD (81.8%) but still fell short of the full model. These results are consistent with the downstream fairness outcomes reported in Table 2: the full model achieves the strongest reductions in centroid distance and MMD, and also the highest Min DI.

Probe accuracy remained relatively stable across all variants (0.70–0.71), compared to 0.72 for the baseline. This

indicates that some sensitive information remains linearly recoverable from the latent representation even after debiasing. However, the large reductions in centroid distance and MMD suggest that the dominant group-level structure has been substantially weakened, even if residual non-linear sensitive signals persist. This is consistent with the design rationale discussed in Section 3.4.8: the objective is to reduce actionable sensitive leakage rather than achieve complete demographic invariance.

Figure 5 provides a qualitative view of the latent space before and after debiasing using PCA. In the baseline representation, male and female candidates show visible separation along the first principal component, with male samples concentrated towards lower values and female samples towards higher values. In the Fair CVAE representation, this separation is substantially reduced, with male and female samples overlapping more uniformly across both principal components.



Fig. 5. PCA visualisation of the latent space, coloured by gender. Left: baseline MLP hidden representation showing visible gender separation along the first principal component. Right: Fair CVAE (full model) latent representation showing reduced gender separation and greater distributional overlap.

To complement the quantitative metrics, kernel density estimates were computed along the first two t-SNE dimensions for both the baseline hidden representation and the Fair CVAE encoder mean μ . Figures 6 and 7 provide a qualitative view of the male and female distributions in the corresponding 2D embeddings.

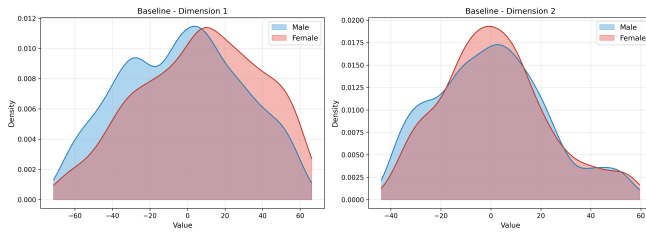


Fig. 6. Kernel density estimates of the 2D t-SNE embedding of the baseline hidden representation, separated by gender.

Because t-SNE is a non-linear projection fitted independently for each representation and is particularly sensitive to residual non-linear structure, these visualisations should be treated as qualitative supplements rather than primary evidence. The reliable evidence for representational change comes from the quantitative metrics in Table 4, which show a 92.4% reduction in centroid distance and a 93.2% reduction in MMD between the baseline and the full Fair CVAE. Together, the quantitative metrics and qualitative

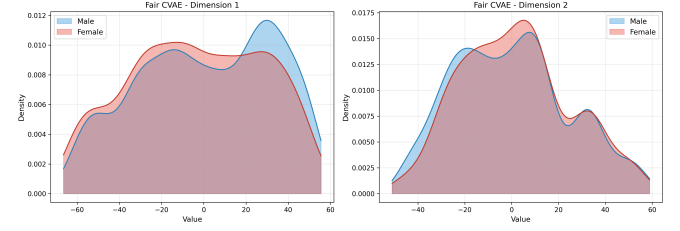


Fig. 7. Kernel density estimates of the 2D t-SNE embedding of the Fair CVAE encoder mean μ , separated by gender.

visualisations support the interpretation that the fairness improvements observed at the prediction level are associated with genuine representational changes in the latent space, rather than being artefacts of threshold calibration.

5 EVALUATION

5.1 Answer to the Research Question

The results provide a positive but qualified answer to the research question. The full Fair CVAE reduced intersectional bias more effectively than standalone adversarial debiasing in the reported experiments, and the latent-space metrics support the claim that this improvement reflected representational change rather than threshold calibration alone. However, the method did not fully satisfy the four-fifths rule for every subgroup, and the evidence is limited by the synthetic dataset and single main random seed. Therefore, the project demonstrates comparative improvement over the selected baselines, but not complete or externally validated fairness.

5.2 Main Result

The main result is that minimum DI increased from 0.34 to 0.70, while accuracy and F1 decreased by approximately three percentage points each. This represents a meaningful fairness improvement at a moderate predictive cost rather than a severe trade-off between fairness and accuracy. At this operating point, the model remains practically useful as a classifier, and the fairness gain cannot be attributed solely to threshold calibration, since both latent centroid distance and MMD decreased by more than 90%. This indicates that the representation itself changed substantially. The remaining gap to the 0.80 threshold is therefore a limitation of the current model, but it does not invalidate the main research aim, which was to assess comparative improvement over the baseline and standalone adversarial methods rather than to guarantee complete threshold satisfaction for every subgroup.

5.3 Interpretation of the Ablation Results

The ablation results explain why the full model worked. Standalone GRL improved minimum DI from 0.34 to around 0.47–0.50, but remained far below the 0.80 threshold. The CVAE *adv_only* variant improved minimum DI further, while *no_adv* achieved stronger latent-space overlap but weaker downstream fairness. Only the full model reached 0.70, indicating that no single mechanism was sufficient on its own.

The *adv_only* result shows that adversarial pressure is useful but incomplete. It improved minimum DI substantially compared with the baseline, but its weaker centroid and MMD reductions indicate that the latent distributions still retained noticeable group structure. This supports the argument that adversarial debiasing alone can suppress direct sensitive predictability without fully reshaping the representation.

The *no_adv* variant is particularly informative because it achieved strong latent-space overlap but weaker downstream fairness. This suggests that reducing marginal distributional differences in the latent space is not sufficient by itself. The predictor may still exploit residual sensitive proxies that remain useful for the hiring task. Therefore, latent-space overlap and decision-level fairness are related but not identical objectives.

The full model performed best because its mechanisms address different failure modes. HSIC reduces broader statistical dependence between the latent representation and the sensitive attribute, the adversary reduces sensitive information that is directly actionable by a classifier, and projection removes dominant sensitive directions in the latent geometry. The ablation results therefore suggest that these mechanisms are complementary rather than interchangeable.

5.4 On the Evaluation Metrics

Using multiple fairness metrics was important, because no single metric provides a complete account of fairness. Disparate Impact is sensitive to the decision threshold, while Equal Opportunity depends on the construction of a proxy qualified subset, since the synthetic dataset does not contain an externally validated merit label. By contrast, the latent-space metrics — centroid distance, probe accuracy, and MMD — are independent of the final decision threshold. Agreement between these latent-space metrics and Disparate Impact therefore strengthens the interpretation that the observed fairness gains are not merely artefacts of threshold calibration. Reporting only one of these metrics would have provided an incomplete basis for evaluation.

Thresholds were calibrated on the validation set rather than the test set. This was a deliberate design choice, because tuning thresholds on the same data used for final reporting would have weakened the validity of the comparison. All reported test-set results therefore use thresholds selected without access to the test labels, helping to preserve a fair evaluation protocol across all models.

The Equal Opportunity results should be interpreted as approximate rather than definitive. Because the dataset is synthetic, there is no ground-truth merit label that can be treated as an externally validated qualification standard. Instead, the qualified subset was constructed from experience, education, algorithmic skill, and interview-related features. This provides a reasonable proxy for comparing opportunity across groups, but it cannot fully resolve the ambiguity inherent in measuring fairness on simulated data.

Disparate Impact also has limitations. Its value depends on the choice of reference group and on the operating threshold used to convert predicted probabilities into binary decisions. For this reason, Disparate Impact alone would not

have been sufficient to support the claims made in this paper. The combination of threshold-dependent group metrics and threshold-independent latent-space metrics provides a more robust basis for assessing whether the proposed method improves fairness in a meaningful way.

5.5 What the Latent Space Shows

Looking inside the latent space was important. If the full model had improved DI without changing the latent representation, the result would provide weaker evidence of genuine debiasing, since the improvement could have been caused primarily by threshold calibration. But centroid distance and MMD both dropped by over 90 percent on the encoder mean, so the male and female distributions in latent space are genuinely much closer than in the baseline. This strengthens the interpretation that the DI improvement reflects representational change rather than threshold effects alone.

Probe accuracy presents a more cautious picture. It changed only marginally, from 0.72 in the baseline to 0.70 in the full model. A logistic probe can still recover gender from the latent space. This is important, because it indicates that the model reduced bias but did not remove it entirely. What the centroid and MMD numbers show is that the dominant, cluster-level gender structure has been broken; what the probe number shows is that finer signals still exist. These findings are therefore complementary rather than contradictory.

The t-SNE plots are less dramatic than the centroid and MMD numbers would suggest. That is a property of t-SNE, not of the model: t-SNE preserves local neighbourhoods at the cost of global distances, so a 90 percent reduction in a 64-dimensional metric does not necessarily become a visible difference in a 2D projection. The t-SNE plots are included for qualitative context. The quantitative claims in this paper rest on the full-dimensional metrics.

5.6 Limitations and Threats to Validity

The results should be interpreted with several validity threats in mind, including internal validity, external validity, construct validity, and statistical conclusion validity. Internally, the final Disparate Impact values depend partly on design choices such as validation-set threshold calibration, adversary reset frequency, and projection update frequency. These choices were fixed for the reported experiments, but alternative settings could change the exact trade-off between fairness and predictive performance.

Externally, the use of synthetic data limits generalisation. The controlled dataset is useful because the injected bias mechanisms are known, but real hiring data may contain unobserved confounders, selection bias, missing data, and institutional effects that are not captured by the benchmark. The results therefore demonstrate effectiveness under controlled conditions rather than direct real-world validity.

Third, there are construct validity limitations in how fairness is defined. Minimum Disparate Impact is useful for measuring worst-case group-level outcome disparity, but it is not a complete definition of fairness. Proxy Equal Opportunity was included as a secondary diagnostic, but

it relies on a proxy qualified subset rather than an externally validated merit label. Therefore, the results should be interpreted as evidence of improved group-level fairness, not as proof that the model satisfies all possible fairness definitions.

Finally, the main results are based on a single seeded run, which limits statistical conclusion validity. The observed ordering between the full model and the simpler baselines is informative, but exact values such as 0.7011 should not be over-interpreted without a multi-seed robustness study.

5.7 Reflection on the Build

The project was built in stages: bias diagnosis, simple baselines, adversarial baselines, then the Fair CVAE. Each stage answered one question and raised the next. Simpler alternatives such as feature removal were considered, but were not adopted as primary mitigation strategies because they risk discarding useful predictive information while leaving proxy bias in correlated variables. The standalone GRL baselines produced measurable but insufficient fairness improvements. These limitations motivated the final Fair CVAE design as a more comprehensive representation-level intervention.

The four-phase training schedule was not a cosmetic choice. Early experiments that turned on the adversary, HSIC, and projection all at once from epoch 1 failed to converge — the predictor never learned anything useful before the fairness pressure destroyed the representation. Phase 1 is there to let the model learn the task first; the fairness constraints are introduced only after something exists to constrain.

A FastAPI web application was built around the same pipeline. While the models constitute the main research contribution, the application provides a reproducible interface for running the full pipeline and a concrete demonstrator of the project, which supports both transparency and future extension.

6 CONCLUSION

This project investigated whether a structured latent representation could reduce bias in automated hiring decisions more effectively than standalone adversarial debiasing. A synthetic hiring dataset was constructed with controlled bias across resume, interview, and culture-fit features. A baseline MLP, two standalone GRL-based adversarial models using gender and intersectional adversarial targets, and three Fair CVAE modes (*adv_only*, *no_adv*, and *full*) were evaluated using predictive metrics, Disparate Impact, and latent-space analysis.

The main finding, discussed in detail in Section 5, is that the full Fair CVAE achieved the strongest observed group-level fairness improvement on this controlled synthetic benchmark: minimum intersectional Disparate Impact rose from 0.34 to 0.70, while accuracy decreased from 0.6903 to 0.6597, and centroid distance and MMD between gender groups in the latent space dropped by more than 90%. Ablation results indicated that no single fairness mechanism was sufficient on its own; the combination of variational bottleneck, HSIC regularisation, adversarial training, and projection was necessary to reach this operating point.

Beyond the technical result, the project contributes a reusable Fair CVAE architecture, a four-phase curriculum training schedule that stabilises joint optimisation across competing objectives, a controlled synthetic hiring benchmark with intersectional bias structure, and a representation-level evaluation protocol that combines threshold-dependent fairness metrics with threshold-independent latent-space metrics. Together, these contributions provide a concrete template for building and evaluating fairness-aware models in hiring-style classification settings.

The findings are also relevant beyond the immediate technical contribution. As recent legislation on automated employment decision tools begins to require independent bias audits and candidate notification, the question of how to actively reduce bias — not merely measure it — has become directly relevant to both practitioners and regulators. The results in this project suggest that naive interventions such as removing the sensitive attribute or applying single-mechanism adversarial debiasing are unlikely to satisfy the four-fifths threshold for the most disadvantaged subgroups, and that more structured representation-level interventions may be necessary in practice.

The method did not eliminate bias completely, as discussed in Section 5: the Female_Black subgroup remained below the four-fifths threshold, residual gender information was still linearly recoverable from the latent space, and the results are limited by the use of synthetic data and a single main random seed. These limitations narrow the strength of conclusions drawn from a single benchmark, but they do not undermine the central finding that the layered Fair CVAE outperforms single-mechanism baselines in the reported experimental setting.

Overall, the project demonstrates that fairness in hiring prediction can be meaningfully improved on this controlled synthetic benchmark by combining several complementary representation-level mechanisms within a single shared latent space. The results support a reduction in dominant sensitive group structure, rather than complete removal of sensitive information, and the resulting fairness gains are supported by genuine changes in the learned representation rather than by threshold effects alone.

6.1 Future Work

Future work should first strengthen the fairness objective itself. The current Fair CVAE adversary primarily targets gender, while fairness is evaluated across gender-race intersectional groups. A natural extension would be to train additional adversaries for race and full intersectional group membership, or to use a multi-adversary architecture that applies fairness pressure at several demographic levels simultaneously. This may help address the remaining disadvantage observed for Black intersectional subgroups, which were the most resistant to mitigation under the current design.

Further work should also improve robustness and generalisation. The experiments should be repeated over multiple random seeds and broader hyperparameter settings, reporting mean and standard deviation for both predictive and fairness metrics, so that the numerical improvements

can be interpreted with greater statistical confidence. The projection mechanism could also be extended beyond linear sensitive directions using non-linear probes or kernel-based subspace removal, which may reduce the residual gender information still recoverable by a linear probe in the current model.

A third direction concerns external validity. The method should be tested on real or semi-realistic hiring datasets, or on standard fairness benchmarks, to assess whether the observed representation-level fairness gains generalise beyond the controlled synthetic setting. Direct comparison against existing fair representation baselines such as the Variational Fair Autoencoder and LAFTR would also place the proposed Fair CVAE within the wider fair representation learning literature, rather than only against simpler adversarial baselines.

Finally, the evaluation could be broadened beyond group-level metrics. Counterfactual fairness analysis [18], in which predictions are compared across hypothetical demographic interventions on otherwise identical candidates, would provide a more individual-level view of fairness that complements the Disparate Impact and latent-space metrics used in this project. Combining representation-level debiasing with causal modelling of the hiring pipeline is a longer-term direction that may help bridge statistical and structural notions of fairness in automated employment decisions.

STATEMENT ON THE USE OF GENERATIVE AI

I used generative AI tools during the preparation of this report to help me understand some concepts more clearly and to improve the grammar, clarity, and wording of my English writing. The project idea, implementation, experiments, results, analysis, and final decisions about the content of the report are my own.

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